

Predicting Stock Price Jumps using Machine Learning

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Keywords

logistic regression, gradient-boosted trees

1 Introduction

Stock prices typically evolve through a combination of continuous fluctuations and occasional abrupt movements. While standard asset pricing models often focus on smooth price dynamics driven by diffusive shocks, financial markets also experience sudden and large price changes, commonly referred to as stock price jumps. These jumps reflect the arrival of new information, changes in investor sentiment, or sudden changes in economic and financial conditions, and play a central role in periods of market stress. Empirically, jumps are associated with increased uncertainty and contribute disproportionately to overall market volatility, making them an important object of study in financial economics.

Understanding the macroeconomic and financial conditions under which stock price jumps occur is crucial for both risk management and asset pricing. Prior research has shown that jump activity tends to increase during recessions, financial crises, and periods of policy uncertainty, suggesting that jumps capture dimensions of risk not fully explained by continuous volatility measures. Motivated by this literature, this report examines the relationship between stock price jump frequency and key macroeconomic and financial variables, including unemployment, monetary policy, inflation, and market-wide volatility.

In addition to documenting these empirical relationships, this report explores the predictability of stock price jumps using a range of machine learning classification models. These models are designed to assess whether observable financial and macroeconomic variables contain information that helps distinguish jump occurrences from non-jump periods. The analysis includes linear models such as logistic regression, as well as regularized variants using LASSO and Ridge penalties, which impose structure to reduce overfitting and improve generalization. To capture potential nonlinear relationships, the study also considers a k-nearest neighbors classifier and a gradient boosting model based on decision trees.

By combining descriptive macro-financial analysis with supervised learning approaches, this project provides a comprehensive view of how stock price jumps co-move with economic conditions and how well different modeling strategies perform in identifying jump episodes. The results shed light on the role of macroeconomic uncertainty in driving jump risk and highlight the strengths and limitations of commonly used machine learning models in this context.

2 Implementation

For this project, data from the CRSP dataset was used, specifically from 1996 to 2024, as well as recession data from the NBER. Data on unemployment, fed funds, VIX, and inflation were also

pulled from the FRED API. The CRSP dataset provides detailed security-level identifiers, price information, trading activity, and return measures used in empirical asset pricing analysis. Each security is uniquely identified by a permanent numerical identifier (PERMNO) and is linked at the firm level through a permanent company identifier (PERMCO). Share codes (SHRCD) classify securities by share type, while industry affiliation is captured through Standard Industrial Classification (SIC) codes. Firm names and trading symbols are provided via company names (COMNAM) and ticker symbols (TICKER), and securities are further referenced using CUSIP identifiers. Daily market information includes transaction prices (PRC), bid and ask quotes (BID, ASK), and daily low and high quote bounds (BIDLO, ASKHI), which together characterize prevailing market valuations and liquidity conditions. Trading activity is measured by daily share volume (VOL) and shares outstanding (SHROUT), allowing for the construction of market capitalization and liquidity proxies. Return measures include raw security returns (RET), returns excluding distributions (RETX), and the value-weighted market return (vwret), which serves as a benchmark for market-wide performance in asset pricing tests. The NBER data provides a monthly snapshot of whether the US economy was in a recession or not, shown as a binary flag with 1 indicating recession and 0 indicating no recession. The FRED data also provides a monthly snapshot of what level or percent a specific macroeconomic indicator was at a certain point in time.

The first step for the project was to clean and merge each dataset. A PreProcessor class was created which could handle all 3 of the datasets. For CRSP, the PreProcessor class removes non-numeric values from a given column, maps SIC codes to specific industries, and ensures consistent datetime formatting for the date column. For the US Recession dataset, the only preprocessing needed is to again ensure consistent datetime formatting. For the data pulled from FRED, I adjusted the CPI from an index to a rate for better interpretability and again ensured consistent datetime formatting.

The next step was to build a variety of features. A FeatureEngineer class was built that would be able to take the processed DataFrame built by PreProcessor and lag returns, check if there is a 10 percent or higher change in value (classified as a jump), build rolling features related to excess returns, volatility, skewness, and beta. A random sample of 100 firms per year is then created. For a better understanding of the data, the Visualizer class looks at the macroeconomic indicators pulled from FRED and compares them to the jump frequency (average number of jumps in a month).

Next, a ModelPipeline class is built that handles the entire pipeline to run the model, including any further preprocessing, fitting and evaluating the model, and then storing results. The 5 models that ModelPipeline can run are a basic Logistic Regression, Logistic Regression with L1 penalty, Logistic Regression with L2 penalty, K-Nearest Neighbors Classifier, and Light GBM Classifier. The Analyzer class can then take the results stored by ModelPipeline and rank them based on metrics such as ROC Score, Accuracy, or the KS Statistic.

The repository for the implementation of this project can be found [here](#).

3 Macroeconomic Conditions and Jump Frequency

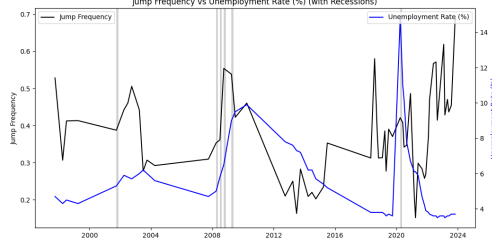


Figure 1: Stock Price Jumps and Unemployment Rate

Figure 1 overlays the unemployment rate from 1996 to 2024 with the frequency of stock price jumps observed in the dataset. During economic downturns, particularly the early 2000s recession, the 2008–2009 global financial crisis, and the 2020 COVID-19 shock, both unemployment and jump frequency increase sharply. This pattern indicates that recessions coincide with heightened equity market volatility.

A notable feature of the relationship is the timing of these movements. Jump frequency often rises before or at the onset of recessions, reflecting financial markets responding immediately to increased uncertainty, while unemployment tends to lag as economic distress spreads through the labor market. This dynamic is especially evident during the 2008 financial crisis, when jump frequency remained elevated for an extended period while unemployment rose steadily and remained high. The COVID-19 recession produced the most dramatic spikes in both series, although the episode was short-lived: unemployment fell rapidly in 2021, while jump frequency normalized but remained somewhat elevated relative to pre-pandemic levels.

Outside of recessions, unemployment follows a gradual downward trend, particularly during the long expansion from 2010 to 2019. Over these periods, jump frequency fluctuates modestly but remains far below crisis levels. Toward the end of the sample, unemployment remains near historic lows, yet jump frequency exhibits periodic spikes, suggesting that financial markets remain sensitive to shocks even in the absence of labor market deterioration.

4 Monetary Policy and Jump Frequency

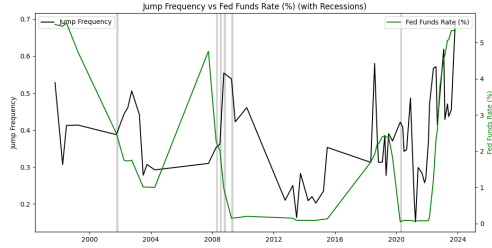


Figure 2: Stock Price Jumps and Fed Funds Rate

The relationship between the Federal Funds Rate and jump frequency highlights the role of monetary policy in shaping market volatility. During periods of monetary tightening, such as the late 1990s and mid-2000s, the Federal Funds Rate rose substantially while jump frequency remained relatively moderate until broader economic stress emerged. Once recessions began, the Federal Reserve responded with aggressive rate cuts, and jump frequency spiked sharply.

This pattern is particularly pronounced during the 2008 financial crisis, when policy rates fell to near zero as jump frequency surged to its highest levels. The COVID-19 recession followed a similar trajectory, with rapid rate cuts in early 2020 coinciding with an extreme but short-lived spike in jump activity. Outside of recessionary periods, extended intervals of low interest rates, such as 2009–2016, are associated with more stable though still noisy jump frequency. More recently, the Federal Reserve’s aggressive rate hikes beginning in 2022 coincided with renewed spikes in jump frequency, suggesting that monetary tightening itself can amplify market volatility even in the absence of a formal recession.

5 Market Volatility and Jump Frequency

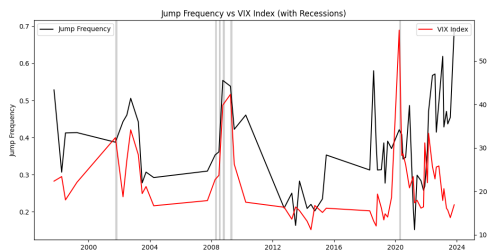


Figure 3: Stock Price Jumps and VIX Index

Figure 3 compares jump frequency with the VIX index, a standard measure of market-wide implied volatility. The two series move closely together, indicating that large individual stock jumps and aggregate market volatility are strongly correlated indicators of financial stress. This relationship is most pronounced during major crises. During the 2008 financial crisis, both jump frequency and the VIX reached extreme levels, reflecting widespread panic and uncertainty. A similar pattern emerged during the COVID-19 recession, when the VIX surged above 60 alongside a sharp increase in jump frequency, though both measures normalized quickly as conditions stabilized.

Outside of recessions, the VIX typically remains at relatively low levels while jump frequency continues to fluctuate modestly. This divergence suggests that while the VIX captures systemic, market-wide volatility, jump frequency may also reflect idiosyncratic firm-level shocks that occur even when broad market volatility is subdued.

6 Inflation and Jump Frequency

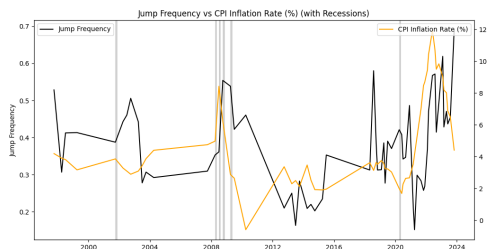


Figure 4: Stock Price Jumps and Consumer Price Index

The relationship between inflation and jump frequency varies across economic regimes. During the 2008 financial crisis, jump frequency spiked sharply while inflation declined due to collapsing demand and falling commodity prices, illustrating how recessions can reduce inflationary pressure

even as financial volatility intensifies. In contrast, the 2021–2022 period featured a surge in inflation to its highest levels in decades, peaking above nine percent, alongside repeated spikes in jump frequency. This pattern suggests that inflation shocks themselves can generate uncertainty and volatility as investors adjust to tighter monetary policy and changing macroeconomic conditions.

Outside of these extreme episodes, inflation and jump frequency often diverge. Much of the 2010s was characterized by historically low and stable inflation, yet jump frequency remained noisy. Similarly, periods of moderate inflation in the late 1990s and early 2000s were not consistently associated with elevated jump activity. Overall, these patterns indicate that inflation alone is not a reliable indicator of increased jump frequency.

7 Model Performance

Table 1 reports performance metrics for the classification models used to predict stock price jump occurrences. Among the models considered, LASSO regression achieves the lowest misclassification rate at 0.3484 and the highest ROC AUC and KS statistics, indicating superior overall performance in both classification accuracy and discriminatory power. These results suggest that imposing an ℓ_1 penalty improves generalization by selecting a sparse set of informative predictors while discarding noise. Logistic regression and Ridge regression exhibit similar performance. Both models produce moderate ROC AUC and KS values but higher misclassification rates relative to LASSO. This pattern indicates that while linear decision boundaries capture meaningful signal in the data, the absence of effective variable selection limits their ability to optimally separate jump and non-jump observations. Ridge regularization improves probability ranking slightly relative to plain logistic regression but does not meaningfully reduce classification error.

LightGBM delivers competitive discrimination metrics, with ROC AUC and KS scores comparable to those of the linear models. However, its misclassification rate remains higher than that of LASSO, suggesting that the additional model flexibility does not translate into superior out-of-sample classification accuracy. This outcome may reflect the presence of noisy predictors, limited nonlinear structure, or class imbalance, which can hinder the effectiveness of tree-based boosting methods in this setting. K-nearest neighbors performs the worst across all metrics, exhibiting the highest misclassification rate and substantially lower ROC AUC and KS statistics. This result is consistent with the high-dimensional and noisy feature space, in which distance-based methods struggle to identify meaningful neighborhoods.

Table 1: Classification model performance metrics

Model	Misclassification Rate	ROC AUC	KS Score
Logistic Regression	0.3803	0.6672	0.2682
LASSO	0.3484	0.6902	0.2981
Ridge	0.3779	0.6686	0.2728
KNN Classification	0.3952	0.5895	0.1375
LightGBM	0.3629	0.6734	0.2729